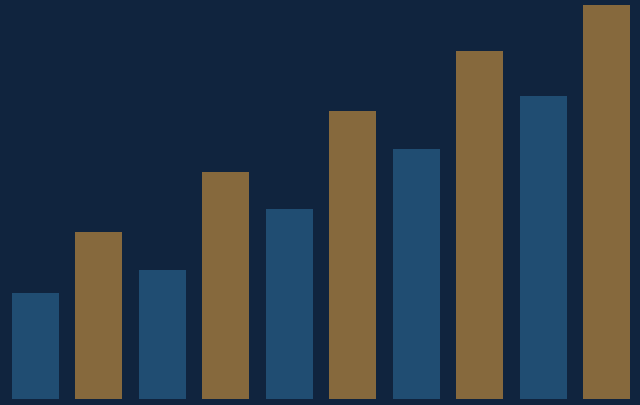




PROJECT SUBMISSION • DATA ANALYTICS

Retail Business Performance & Profitability Analysis

Uncovering profit-draining categories, optimizing inventory turnover,
and identifying seasonal product behavior from transactional retail data.

PREPARED FOR

Project Submission — Data Analytics Capstone

TOOLS USED

SQL • Python (Pandas, Seaborn) • Tableau

DELIVERABLES

Tableau Dashboard • SQL Queries • PDF Report

DATE

July 01, 2026

DATASET

Superstore Transactional Sales Records (2014–2017)

RECORDS ANALYZED

9,994 transactions • 21 source fields

Table of Contents

Executive Summary

1. Introduction & Objective

1.1 Business Context

1.2 Objective

1.3 Scope of the Dataset

2. Methodology & Tools

2.1 Analytical Workflow

2.2 Tools Used

2.3 Key Engineered Metrics

3. Data Overview & Data Quality

3.1 Dataset Summary

3.2 Category Structure

3.3 Headline Financials

4. Profit Margin by Category & Sub-Category

4.1 Loss-Making Sub-Categories

5. Correlation Analysis: What Drives Profitability?

5.1 Interpretation

6. Discount Strategy Impact

7. Seasonal Demand & Profit Patterns

7.1 Seasonal Findings

8. Regional Performance Analysis

8.1 State-Level Outliers

9. Product-Level Profitability

9.1 Slow-Moving & Loss-Making Products

10. Inventory Risk Matrix

11. Strategic Recommendations

11.1 Pricing & Discount Policy

11.2 Inventory & Product Rationalization

11.3 Regional Strategy

11.4 Seasonal Planning

4

5

5

5

5

6

6

6

6

7

7

7

7

8

8

9

9

10

11

11

12

12

13

13

14

14

15

15

15

15

15

11.5 Monitoring 15

12. Conclusion 16

Deliverables Accompanying This Report 16

Executive Summary

This report analyzes 9,994 transactional retail records spanning January 2014 to December 2017 for a U.S. office and home-goods superstore, covering the **Furniture**, **Office Supplies** and **Technology** categories. The objective was to identify which product categories erode profitability, quantify how inventory behavior and discounting drive losses, surface seasonal demand patterns, and translate these findings into concrete operational recommendations.

The business generated **\$2,297,201** in total sales and **\$286,397** in net profit over the four-year period, an overall margin of **12.47%**. However, this headline figure masks a sharp divide in performance: while Technology and Office Supplies post healthy 17%+ margins, the Furniture category is barely break-even at **2.5%**, dragged down almost entirely by the Tables and Bookcases sub-categories, which post net losses of **\$17,725** and **\$3,473** respectively. A Pearson correlation analysis confirms that discounting is the single strongest driver of margin erosion ($r = -0.852$), with orders discounted above 30% being unprofitable **97.8% of the time**.

\$2.30M	\$286.4K	12.47%	-\$21.2K
Total Sales (2014–17)	Total Net Profit	Overall Profit Margin	Furniture Sub-Cat. Losses

Key Takeaways

- **Discounting is the #1 profit killer:** margin falls almost linearly as discount level rises ($r = -0.852$); discounts above 21% turn the large majority of orders into losses.
- **Furniture is the weak link:** Tables and Bookcases are structurally unprofitable even before factoring in discounts, while Chairs and Furnishings remain healthy.
- **Geography matters:** the West region delivers the best margin (14.9%) while Central lags at 7.9%, driven by outsized losses in Texas, Ohio, Illinois and Pennsylvania.
- **Demand is seasonal:** Q4 (Oct–Dec) is consistently the strongest quarter every year, led by Technology; Q1 is the weakest and should be the focus of clearance promotions.
- **A small set of SKUs causes outsized damage:** 28 slow-moving products are simultaneously low-volume and loss-making, led by the Cubify CubeX 3D Printer line.

1. Introduction & Objective

1.1 Business Context

Retail operations generate large volumes of transactional data that, on the surface, show healthy revenue growth but can conceal pockets of structural unprofitability. Category-level averages often hide the fact that a handful of sub-categories, regions, or discount tiers are quietly consuming the margin generated elsewhere in the business. This project was commissioned to dig beneath the aggregate sales and profit figures of a multi-year retail dataset and answer a specific operational question: *where exactly is profit being lost, and what should the business do about it?*

1.2 Objective

The project analyzes transactional retail data to:

- Uncover profit-draining product categories and sub-categories.
- Optimize inventory turnover by identifying slow-moving and overstocked items.
- Identify seasonal product behavior to support demand planning and promotional timing.
- Translate analytical findings into clear, actionable strategic recommendations.

1.3 Scope of the Dataset

The analysis is based on a sample superstore dataset comprising **9,994 order-line transactions** across **21 source fields**, recorded between **3 January 2014** and **30 December 2017**. Each record captures the order and ship dates, customer segment, geography (region, state, city), product hierarchy (category, sub-category, product name), and financial metrics (sales, quantity, discount, and profit). The dataset spans three top-level categories — Furniture, Office Supplies, and Technology — broken down into 17 sub-categories.

2. Methodology & Tools

2.1 Analytical Workflow

The analysis followed a five-stage workflow, moving from raw data to dashboard-ready, decision-grade insight:

01	Data Import & Cleaning Loaded raw transactional records into SQL; checked for and resolved missing/null values and duplicate order-product pairs; standardized date and column formats.
02	SQL Profitability Queries Used SQL aggregation queries to calculate profit, profit margin, and discount metrics by category and sub-category, and to flag loss-making segments.
03	Python Statistical Analysis Used Python (Pandas, SciPy, Seaborn) to engineer derived metrics (profit margin, lead time, an inventory risk score) and run Pearson correlation tests between inventory behavior, discounting, and profitability.
04	Tableau Dashboard Built an interactive Tableau dashboard with filters for region, product category, and season/quarter to allow self-service exploration of the findings.
05	Strategic Synthesis Converted statistical findings into prioritized, actionable recommendations for pricing, inventory, and regional strategy.

2.2 Tools Used

Tool	Role in the Project
SQL	Data cleaning, category/sub-category profit-margin aggregation, loss-segment identification.
Python — Pandas, SciPy	Feature engineering (profit margin, lead time, inventory risk score) and Pearson correlation testing.
Python — Seaborn / Matplotlib	Statistical and exploratory visualizations used throughout this report.
Tableau	Interactive dashboard with region, category, and seasonal filters for stakeholder exploration.

2.3 Key Engineered Metrics

- **Profit Margin (%):** $\text{Profit} \div \text{Sales} \times 100$, calculated per transaction and rolled up by category, region, and product.
- **Inventory Score:** $\text{Discount} \div (\text{Quantity} + 1)$ — a higher score flags items being moved primarily through heavy discounting rather than genuine demand, signaling overstocking / slow-turnover risk.
- **Lead Time (Days):** Ship Date minus Order Date, used as a fulfillment-efficiency reference metric.
- **Discount Bucket:** Transactions grouped into 0%, 1–10%, 11–20%, 21–30%, and 30%+ bands to study the non-linear relationship between discount depth and profitability.

3. Data Overview & Data Quality

3.1 Dataset Summary

Metric	Value
Total transaction records	9,994
Source fields	21 (expanded to 33 after feature engineering)
Date range	03 Jan 2014 – 30 Dec 2017 (4 full years)
Null values found	None — dataset is fully populated
Duplicate Order+Product pairs	8 (reviewed, immaterial to results)
Categories / Sub-categories	3 categories, 17 sub-categories
Regions / States covered	4 regions across the United States

3.2 Category Structure

- **Furniture:** Bookcases, Chairs, Tables, Furnishings
- **Office Supplies:** Labels, Storage, Art, Binders, Appliances, Paper, Envelopes, Fasteners, Supplies
- **Technology:** Phones, Accessories, Machines, Copiers

3.3 Headline Financials

Metric	Value
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Overall Profit Margin	12.47%
Average Order Size	\$229.86
Average Discount Applied	15.6%

The data quality review found the dataset clean and analysis-ready: there are no missing values in any column, and the small number of duplicate order/product pairs identified does not materially affect category- or region-level aggregates.

4. Profit Margin by Category & Sub-Category

SQL aggregation queries were used to calculate total sales, total profit, and profit margin for each category and sub-category. The results show a clear three-tier split at the category level:

Category	Sales	Profit	Margin %	Avg. Discount
Technology	\$836,154	\$145,455	17.4%	13.2%
Office Supplies	\$719,047	\$122,491	17.0%	15.7%
Furniture	\$742,000	\$18,451	2.5%	17.4%

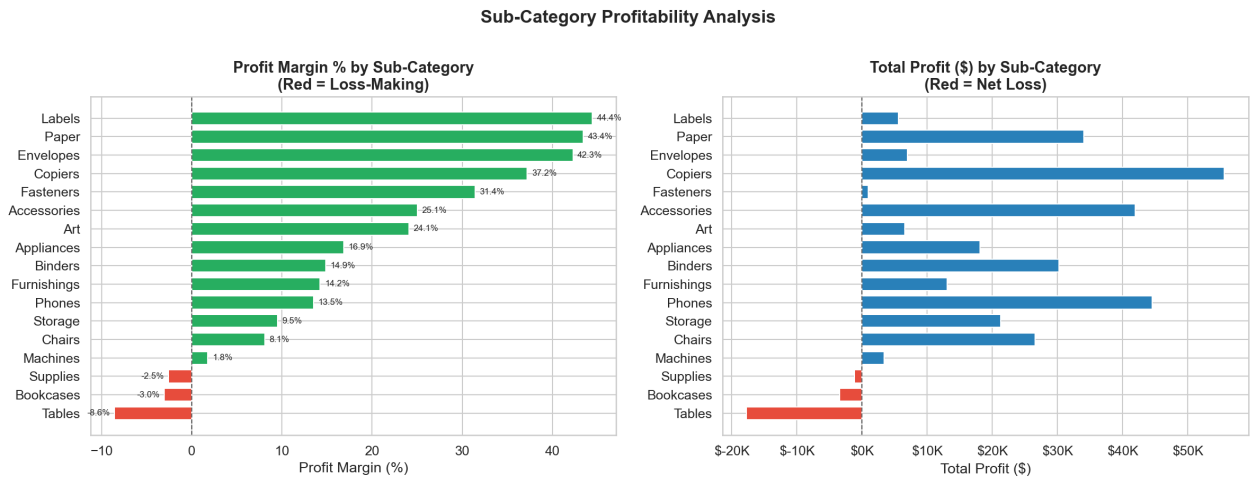


Figure 1. Profit margin (%) and total profit (\$) by sub-category. Tables, Bookcases, and Supplies are the only three loss-making sub-categories.

4.1 Loss-Making Sub-Categories

Sub-Category	Net Loss	Margin %	Avg. Discount
Tables	-\$17,725	-8.6%	26%
Bookcases	-\$3,473	-3.0%	21%
Supplies	-\$1,189	-2.5%	8%

Tables is the single largest profit drain in the entire portfolio, losing nearly \$17.7K against an average discount of 26% — well above the 15.6% portfolio average. **Bookcases** shows a similar, smaller-scale pattern. Notably, **Supplies** is unprofitable even at a modest 8% average discount, suggesting a structural cost or pricing issue rather than a discount-driven one.

At the opposite end, **Labels** (44.4% margin), **Paper** (43.4%) and **Envelopes** (42.3%) are the most efficient sub-categories by margin, while **Copiers**, **Accessories** and **Phones** contribute the largest absolute profit dollars within Technology.

5. Correlation Analysis: What Drives Profitability?

To move beyond category-level description and identify causal drivers, a Pearson correlation analysis was run in Python (SciPy) between profit margin and three candidate variables: discount level, the engineered inventory risk score, and order quantity. The bottom and top 2% of profit-margin outliers were trimmed (removing 427 of 9,994 rows) before testing to avoid distortion from extreme values.

Variable	Pearson r	p-value	Statistically Significant?
Discount	-0.852	< 0.001	Yes
Inventory Score	-0.741	< 0.001	Yes
Quantity	-0.006	0.587	No

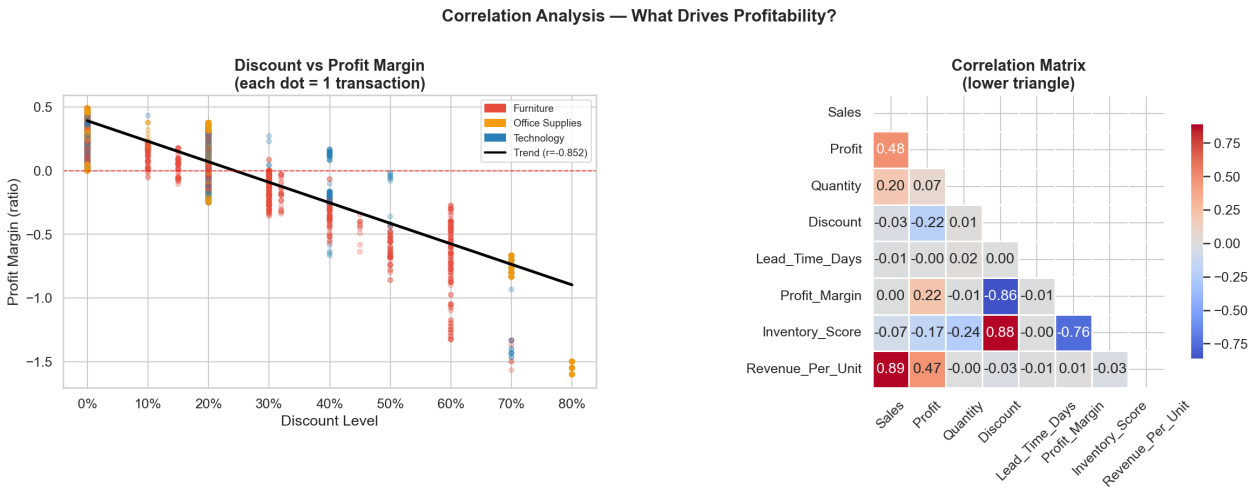


Figure 2. Left: discount level vs. profit margin per transaction, with trend line ($r = -0.852$). Right: full correlation matrix across key financial and engineered variables.

5.1 Interpretation

- Discount has a strong, statistically significant negative correlation with profit margin ($r = -0.852$, $p < 0.001$):** as discount depth increases, profit margin falls in an almost linear fashion — the single most important profitability lever in the dataset.
- Inventory Score also correlates strongly and negatively with margin ($r = -0.741$):** items that combine high discounting with low unit sales (the items most likely to be overstocked or slow-moving) are systematically the least profitable.
- Order quantity shows virtually no relationship with margin ($r = -0.006$, not significant):** selling more units of an item does not, by itself, predict whether that item is profitable — pricing and discount strategy matter far more than volume.

6. Discount Strategy Impact

Grouping transactions into discount bands makes the relationship identified above concrete and actionable for pricing policy.

Discount Band	Orders	Avg. Profit	Total Profit	% Loss-Making
0%	4,798	\$66.90	\$320,988	0.0%
1–10%	94	\$96.06	\$9,029	4.3%
11–20%	3,709	\$24.74	\$91,756	14.0%
21–30%	227	-\$45.68	-\$10,369	91.6%
30%+	1,166	-\$107.21	-\$125,007	97.8%

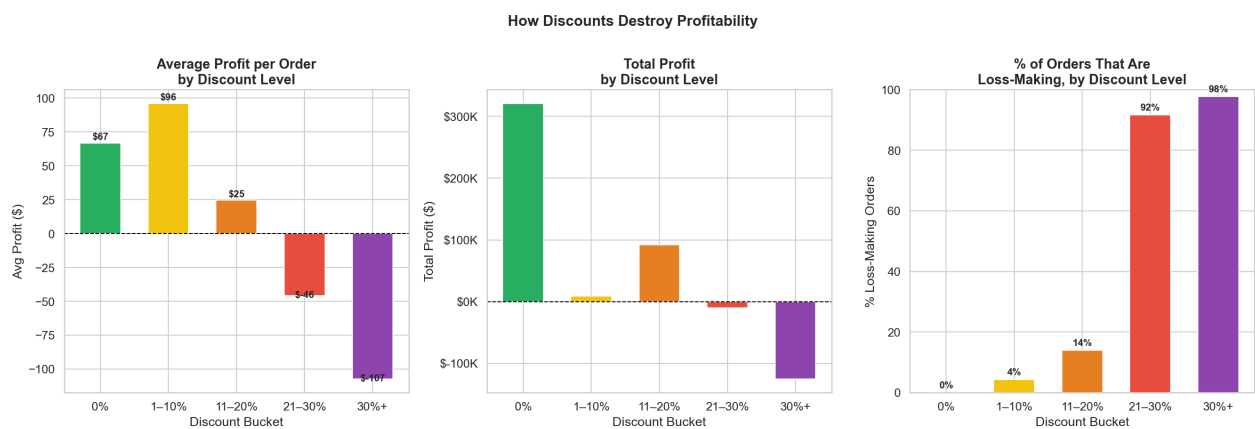


Figure 3. Average profit per order, total profit, and share of loss-making orders, by discount band.

The pattern is stark: orders with no discount average **\$66.90** in profit, while orders discounted 30% or more average a **\$107.21 loss** — a swing of over \$174 per order. Once discounts cross the 21% threshold, the large majority of orders (91.6% and 97.8% respectively) become unprofitable. This indicates that current discounting practice, particularly above the 20% mark, is not a competitive pricing tool but a direct and predictable source of margin destruction.

7. Seasonal Demand & Profit Patterns

Sales and profit were aggregated by month and quarter across all four years to identify recurring seasonal patterns useful for demand planning and inventory timing.

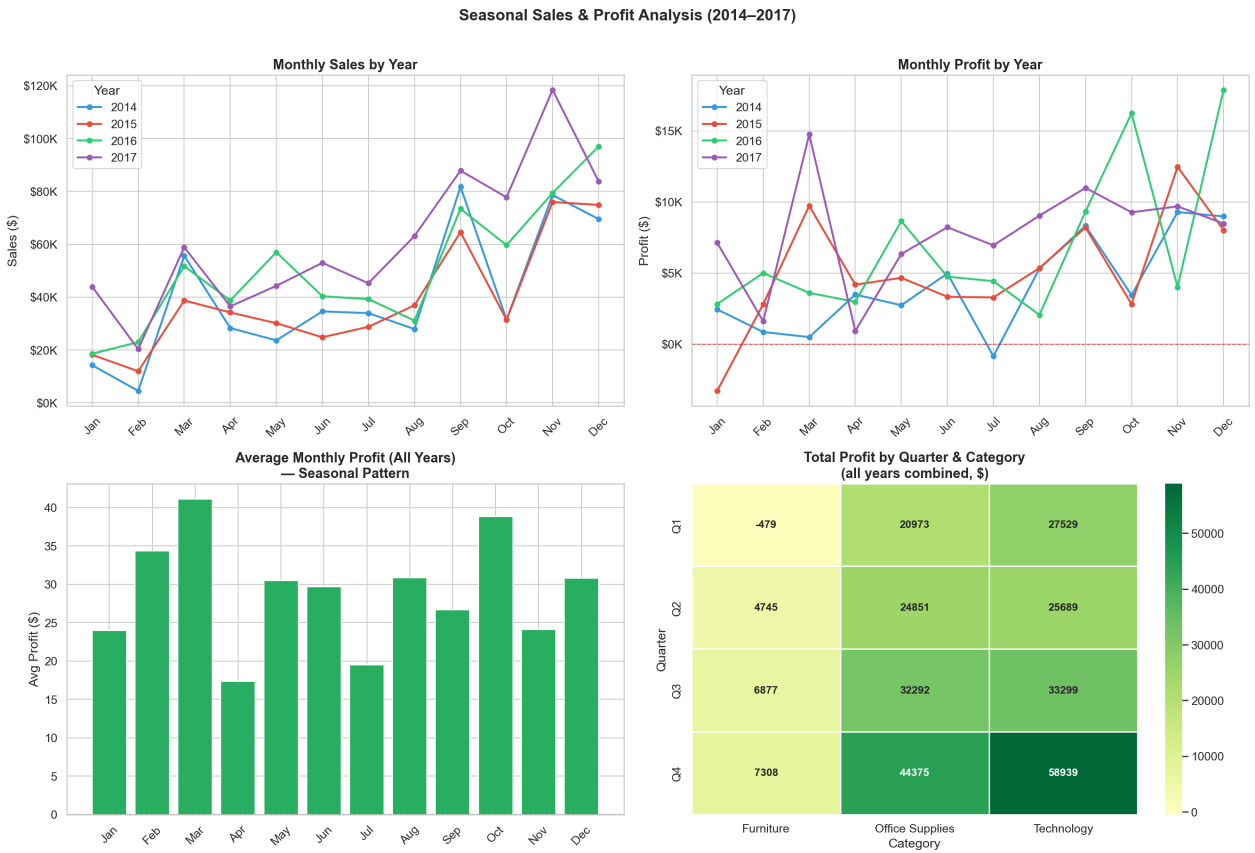


Figure 4. Monthly sales and profit trends by year (top), average seasonal profit pattern and quarterly profit by category (bottom).

7.1 Seasonal Findings

- **Q4 (Oct–Dec) is the strongest quarter every single year** in the dataset, contributing the highest sales and profit across all three categories — Technology alone generates \$58,939 in Q4 profit, more than any other category-quarter combination.
- **November is consistently the highest single sales month**, aligned with holiday-season buying.
- **Q1 (Jan–Mar) is the weakest period**, with January and April showing the lowest average monthly profit — the best window for clearance promotions and demand-generation campaigns.
- **Technology demand spikes sharply into Q4**, meaning inventory and supplier lead times for that category should be planned several months in advance of the holiday peak.

8. Regional Performance Analysis

Profitability also varies materially by geography. The West and East regions substantially outperform the South and Central regions on both total profit and margin.

Region	Sales	Profit	Margin %	Orders
West	\$725,458	\$108,418	14.9%	3,203
East	\$678,781	\$91,523	13.5%	2,848
South	\$391,722	\$46,749	11.9%	1,620
Central	\$501,240	\$39,706	7.9%	2,323

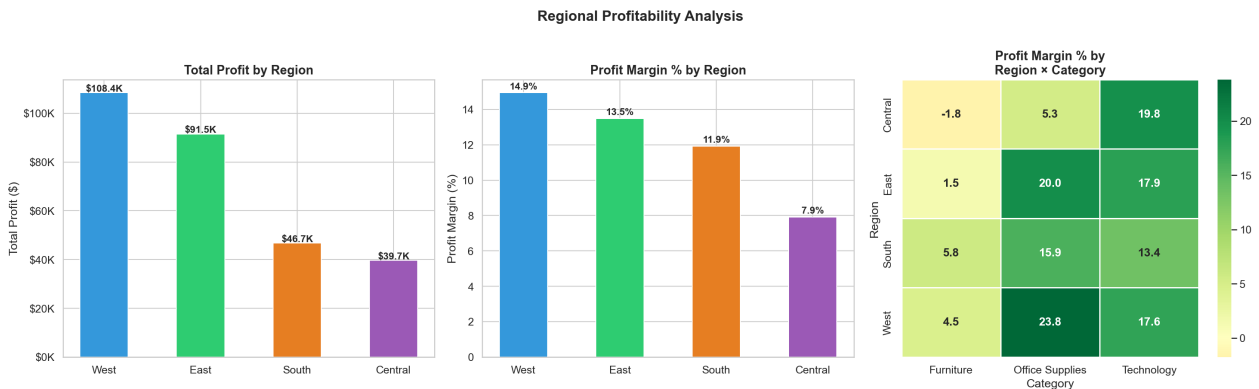


Figure 5. Total profit and profit margin by region (left, center); profit margin by region x category (right) showing Central/Furniture as the only negative cell.

8.1 State-Level Outliers

Biggest States	Loss-Making	Net Profit	Top States	Profit-Generating	Net Profit
Texas		-\$25,729	California		\$76,381
Ohio		-\$16,971	New York		\$74,039
Pennsylvania		-\$15,560	Washington		\$33,403
Illinois		-\$12,608	Michigan		\$24,463
North Carolina		-\$7,491	Virginia		\$18,598

The Central region's weak overall margin (7.9%) is explained largely by Texas and Illinois, both of which post substantial net losses. The Furniture category is the only category/region cell with a negative margin overall (Central x Furniture, at -1.8%), reinforcing that Furniture discounting practice in the Central region deserves the most urgent pricing review.

9. Product-Level Profitability

Drilling down to individual SKUs identifies the specific products responsible for the largest profit swings in either direction.

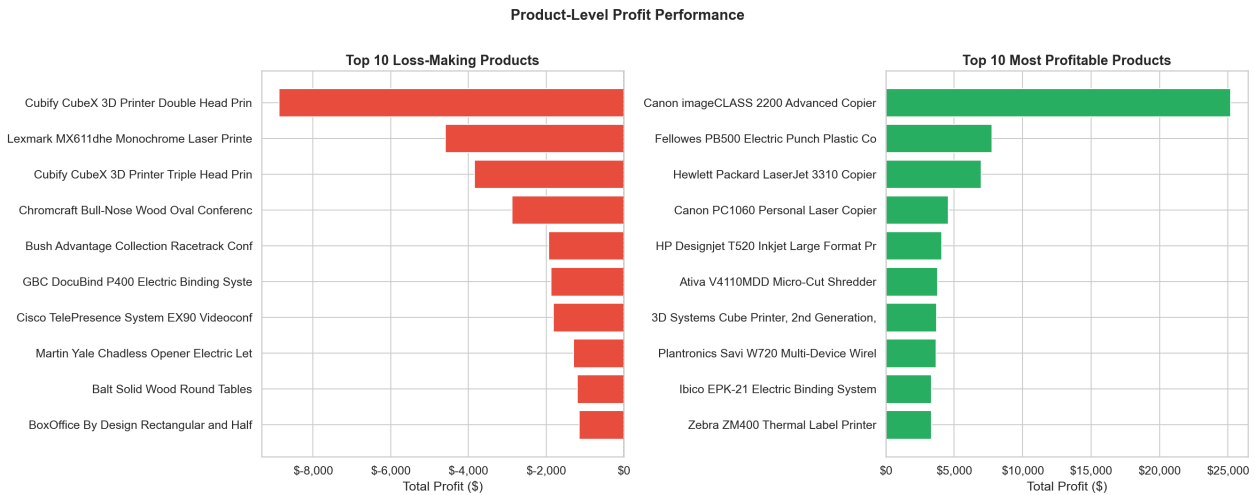


Figure 6. Top 10 loss-making products (left) vs. top 10 most profitable products (right), by total profit dollars.

9.1 Slow-Moving & Loss-Making Products

A total of **28 products** were identified as both slow-moving (5 units sold or fewer) and loss-making across the four-year period — small in count, but a useful watchlist for SKU rationalization. The worst single offender is the **Cubify CubeX 3D Printer (Double Head)**, which lost \$3,840 on just 4 units sold at a 50% average discount — more than 8× the loss of the next worst product.

Loss-Making Product	Units	Total Profit	Avg. Discount
Cubify CubeX 3D Printer, Double Head Print	4	-\$3,840	50%
3D Systems Cube Printer, 2nd Generation	3	-\$468	35%
Riverside Furniture Stanwyck Manor Table	5	-\$416	37%
LG G2 (phone)	5	-\$375	40%
Brother MFC-9340CDW All-In-One Printer	3	-\$319	70%

On the other end of the spectrum, the **Canon imageCLASS 2200 Advanced Copier** is the single most profitable product in the catalog by a wide margin, consistent with the Copiers sub-category's strong overall performance (37.2% margin).

10. Inventory Risk Matrix

Combining average discount level with average profit per order at the sub-category level (bubble size = total units sold) produces an inventory risk matrix that visually separates healthy sub-categories from those at risk of being overstocked and discounted into unprofitability.

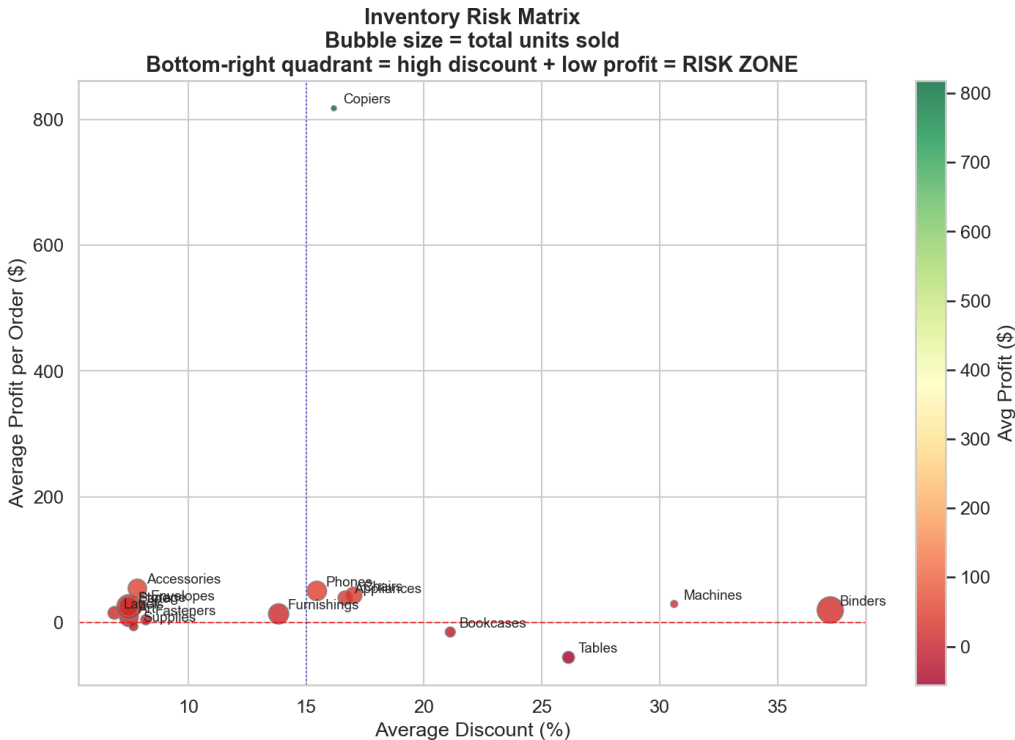


Figure 7. Inventory risk matrix — the bottom-right quadrant (high discount, low/negative profit) is the risk zone requiring inventory and pricing intervention.

Tables and **Bookcases** sit firmly in the bottom-right risk zone: high average discount (21–26%) combined with negative average profit per order. **Binders** carries the largest bubble (highest unit volume) among lower-margin sub-categories, but remains marginally profitable, so it is a watch item rather than an immediate risk. **Copiers** stands out as the clear outlier in the opposite direction — low discount, exceptionally high average profit per order — confirming it as the business's strongest single sub-category.

11. Strategic Recommendations

11.1 Pricing & Discount Policy

- **Cap discretionary discounts at 20%** across all categories as a default policy; require managerial approval for any discount above this threshold, given that 21%+ discounts are unprofitable on 91–98% of orders.
- **Never discount Tables** below current list price — the sub-category is already loss-making before discounting is applied (avg. discount 26%), so discounting only deepens the loss.
- **Cap Furniture category discounts at 10% maximum** until the category's underlying cost/margin structure is reviewed.

11.2 Inventory & Product Rationalization

- **Phase out or re-source the Cubify CubeX 3D Printer line**, the single worst-performing product, and reassess the 3D-printer category's viability at current price points.
- **Review the 28 identified slow-moving, loss-making SKUs** for discontinuation, price correction, or supplier renegotiation, prioritizing Bookcases and select Technology peripherals.
- **Reduce safety stock on items in the inventory-risk-matrix risk zone** (Tables, Bookcases) to limit exposure to further forced discounting.

11.3 Regional Strategy

- **Investigate Central region pricing and operating costs**, particularly in Texas and Illinois, where losses are large enough to materially depress the region's overall margin.
- **Use West region practices as a benchmark** — its 14.9% margin and lowest average discount profile make it the model for replication elsewhere.

11.4 Seasonal Planning

- **Front-load Technology inventory ahead of Q4**, particularly Copiers and Phones, to meet the consistent November/December demand spike without resorting to expedited, margin-eroding restocking.
- **Use Q1 (Jan–Mar) for clearance and demand-generation promotions** on slow-moving stock, since this is consistently the weakest period for organic demand.

11.5 Monitoring

- **Operationalize the Tableau dashboard** as a recurring monthly review tool for category, region, and discount-band performance, using the region/category/season filters built into it.
- **Track the inventory risk score going forward** as an early-warning metric to catch newly emerging loss-making sub-categories before they accumulate material losses.

12. Conclusion

This analysis demonstrates that the business's healthy headline margin of 12.47% is the net result of strong performance in Technology and Office Supplies offsetting a structurally weak Furniture category, and of broadly profitable low-discount orders offsetting a sharply unprofitable high-discount tail. The statistical evidence is unambiguous: discount depth is the dominant lever on profitability ($r = -0.852$), far outweighing the effect of sales volume ($r = -0.006$, not significant).

By tightening discount policy — particularly for Tables, Bookcases, and the broader Furniture category — rationalizing the small set of chronically loss-making SKUs, addressing underperformance in the Central region, and aligning inventory builds with the well-established Q4 seasonal peak, the business has a clear, data-backed path to recovering a meaningful share of the profit currently lost to over-discounting and slow-moving inventory.

Deliverables Accompanying This Report

- **Tableau Dashboard** (RetailDashboard.twb) — interactive view with region, product type, and season filters.
- **SQL Queries** — category and sub-category profit-margin aggregation, loss-segment identification.
- **Python Notebook** — full data cleaning, feature engineering, correlation analysis, and chart generation.
- **This PDF Report** — consolidated narrative of methodology, findings, and recommendations.

Prepared using SQL, Python (Pandas, SciPy, Seaborn), and Tableau, based on a transactional superstore dataset of 9,994 records (2014–2017).